

Coding & Solution

Date	26 June 2026
Team ID	SWTID-2026-6095
Project Name	EduGenie
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/BhavyaAICE/EduGenie
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	FastAPI
Key Features Implemented	1. Concept Explainer Engine: Developed a local AI text generation pipeline using the LaMini-Flan-T5-248M model to translate complex educational topics into simplified terms. 2. Text Summarization System: Integrated the cloud-based Google Gemini Flash API with asynchronous handling (generate_content_async) and precise token thresholds to extract concise, bulleted revision summaries. 3. QnA and Quiz Module Integration: Additional evaluation tracks. 4. Dynamic UI Binding: Connecting the static HTML/CSS frontend templates with the live FastAPI endpoints using JavaScript fetch requests.
Pending / Incomplete Features	NA
Setup / Run Instructions	-BACKEND SETUP 1. Navigate to the backend directory: cd backend

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| | <ol style="list-style-type: none">2. Initialize and activate the virtual environment: <code>python -m venv .venv</code>
&& <code>source .venv/bin/activate</code>3. Install all dependencies: <code>pip install -r requirements.txt</code>4. Configure your <code>.env</code> file with a valid <code>GEMINI_API_KEY</code>.5. Launch the application server: <code>uvicorn app.main:app --reload</code> |
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-FRONTEND SETUP

1. Open the frontend directory containing the HTML and CSS assets.
2. Launch the `index.html` file directly in any standard web browser or via a local live server.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	
2	Meaningful variable and function names are used	
3	Code includes comments / documentation where necessary	
4	Error handling is implemented for critical operations	
5	The application runs without critical errors	
6	Code is committed to a version control repository	

Additional Notes / Comments:

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